ABHINAV GOYAL

Lead Business Analyst · E-commerce & FMCG Analytics

"Cut the Noise. Own the Edge."

IIT Kanpur · Noon.com · Gurugram, India · July 2026

avi290994@gmail.com · agdecodes.github.io · linkedin.com/in/abhinav-goyal-7814105a

7+	39+	6	12	2235
Years Experience	Projects Delivered	Markets Covered	AMs Supported	IIT-JEE AIR

Portfolio: agdecodes.github.io · LinkedIn: abhinav-goyal-7814105a · GitHub: agdecodes

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01 · ABOUT ABHINAV — IDENTITY, VALUES & PHILOSOPHY

Who Is Abhinav Goyal?

Abhinav Goyal (AG) is a data-driven business analyst with 7+ years of experience turning raw transactional data into decisions that move commercial levers. Currently operating as Lead Business Analyst at Noon.com — MENA's largest e-commerce platform — he owns end-to-end analytics for KSA Beauty and AE/SA FMCG categories, directly supporting 12 Account Managers across Saudi Arabia and the UAE. His work spans production BigQuery pipeline engineering, CEO-level Excel decks, and WhatsApp CRM cohort targeting. He is equally comfortable writing 500-line SQL and presenting a 7-tab business intelligence deck to senior leadership.

Abhinav's edge lies in bridging the gap between raw data infrastructure and business action. He doesn't just analyse — he builds systems, automates pipelines, and engineers analytical frameworks that scale across teams and quarters. His tagline, 'Cut the Noise. Own the Edge,' reflects a philosophy of ruthless prioritisation and high-signal output.

Current Role	Lead Business Analyst, Noon.com (Feb 2025 – Present)
Markets	Saudi Arabia (KSA), United Arab Emirates (UAE)
Categories	KSA Beauty, AE/SA FMCG (Grocery, Baby, Hair & Personal Care)
Location	Gurugram, India (Remote)
Email	avi290994@gmail.com
Portfolio	agdecodes.github.io
LinkedIn	linkedin.com/in/abhinav-goyal-7814105a
GitHub	github.com/agdecodes

Core Values & Working Philosophy

- Data without action is noise — every analysis ends with a clear recommendation.
- Ship production-ready code, not POCs. If it goes into a pipeline, it must be bulletproof.
- Business context before SQL. Understand what the stakeholder needs before touching a keyboard.
- Extreme ownership — if a query is wrong, the downstream business decision is wrong too.
- Communicate in the language of the business: GMV, Units, ASP, OOS %, Price Competitiveness.
- Automate repetitive work relentlessly — if you run a query more than twice, it should be a pipeline.

02 · EDUCATION — IIT KANPUR & ACADEMIC ACHIEVEMENTS

Indian Institute of Technology, Kanpur (IIT-K)

B.S.-M.S. Economics | Minor in Industrial & Management Engineering (IME) | 2013–2018

IIT Kanpur is among the most prestigious engineering and research universities in Asia. The five-year dual degree combines rigorous quantitative training (econometrics, statistics, game theory, micro/macroeconomics) with operations research, financial modelling, and management science.

2,235	113th	B'84 Scholar	5 Years	Economics + IME
IIT-JEE AIR Rank	IMO National Rank	Merit Scholarship	Dual Degree	Major + Minor

- IIT-JEE AIR 2235 — Cleared one of the world's most competitive entrance exams (top 0.2% of ~1.5M candidates).
- International Mathematical Olympiad (IMO) — 113th National Rank — among the top mathematical talent in India at school level.
- Batch of 1984 Scholarship — Prestigious merit scholarship awarded by IIT Kanpur alumni to top-performing students.
- Minor in IME — Industrial & Management Engineering: operations research, supply chain optimisation, decision science.
- Quantitative Core — Econometrics, Statistics, Game Theory provided the intellectual scaffolding for a data career.

03 · CAREER TIMELINE — SIX ROLES ACROSS EIGHT YEARS

Period	Role	Company	Domain	Key Impact
Feb 2025–Present	Lead Business Analyst	Noon.com	E-com / Beauty / FMCG	KSA Beauty + AE/SA FMCG; 12 AMs; Produ
Nov 2022–Feb 2025	Freelance Data Analyst	Self-Employed	Cross-industry	Multi-client dashboards, SQL, automation pip
Jan 2022–Oct 2022	Senior Product Analyst	Eucloid	B2B SaaS / Product	Funnel analytics, retention, A/B testing infra
Aug 2020–Jan 2022	Data Scientist	UnitedHealth Group	Healthcare / Insurance	Claims prediction, NLP, risk stratification
Jul 2018–Aug 2020	Business Analyst	Citibank	BFSI / Banking	Credit analytics, portfolio risk, regulatory repo
2017 (Internship)	BA Intern	Travel Triangle	Travel / D2C	Growth analytics, funnel analysis, cohort trac

Lead Business Analyst — Noon.com (Feb 2025 – Present)

MENA's largest e-commerce platform. Abhinav single-handedly owns the analytics stack for KSA Beauty and AE/SA FMCG. His work directly informs category strategy, AM targets, inventory decisions, and supplier negotiations. He has built 10+ production BigQuery/Airflow pipelines, 15+ SQL workstreams, and multiple CEO-facing Excel decks.

Freelance Data Analyst — Self-Employed (Nov 2022 – Feb 2025)

Multi-client engagements across e-commerce, FMCG, fintech, and healthcare. Built dashboards, automated reporting pipelines, and SQL-based analytics systems for SME and mid-market clients. Deepened BigQuery, Looker Studio, and Apps Script expertise across diverse business contexts.

Data Scientist — UnitedHealth Group (Aug 2020 – Jan 2022)

Healthcare claims prediction, NLP-based document classification, and risk stratification models. Used Python (pandas, sklearn, spaCy), SQL, and Tableau in a large-scale enterprise data environment.

Business Analyst — Citibank (Jul 2018 – Aug 2020)

Credit portfolio analytics, regulatory reporting, and risk dashboarding in a structured BFSI environment. Built the foundation in SQL, Excel modelling, and stakeholder communication with senior banking professionals.

04 · TECHNICAL STACK & TOOLS

Query & Data Engineering

BigQuery SQL (Advanced)	Google Cloud Platform	Airflow DAGs	CTE Architecture
Window Functions	ARRAY_AGG Deduplication	SAFE_DIVIDE	GROUP BY ALL

Dashboarding & Visualisation

Looker Studio	Google Sheets (Advanced)	Apps Script	Excel (Multi-tab, Dynamic Charts)
Pivot Tables	Conditional Formatting		

Programming & Scripting

Python (pandas, matplotlib, reportlab)	Node.js	Apps Script (JS)	Bash Scripting

Analytics & Data Science

Cohort Analysis	Funnel Analytics	A/B Testing	Customer Segmentation
NLP (spaCy, sklearn)	Time-Series Analysis	Regression Modelling	K-means Clustering

AI & Automation

Anthropic Claude API	Railway (Deployment)	LangGraph (Learning)	Hugging Face
Prompt Engineering	LinkedIn Automation Pipeline		

Domain Analytics

Category P&L;	GMV Forecasting	OOS & Pricing	Seller Analytics
CRM & Cohort Targeting	AP/AR Aging	AOP Planning	Price Competitiveness

05 · PERSONAL BRAND — AG ANALYTICS & THE 1% INSIGHT

AG Analytics

Abhinav is building a personal brand called AG Analytics with the tagline 'Cut the Noise. Own the Edge.' The brand is rooted in his belief that the most valuable analysts are not those who produce the most data, but those who extract the highest-signal insights from it. Portfolio: agdecodes.github.io

The 1% Insight — LinkedIn Content Engine

A fully automated LinkedIn content engine publishing twice daily — 10 AM IST (analytical, data-heavy) and 7 PM IST (lighter, viral) — built on Node.js + Anthropic Claude API, deployed on Railway.

Stack	Node.js + Anthropic Claude API + Railway
Schedule	10 AM IST (analytical) + 7 PM IST (viral) — twice daily
LinkedIn	linkedin.com/in/abhinav-goyal-7814105a
Company Page	linkedin.com/company/the-1-insight
GitHub	github.com/agdecodes
Portfolio	agdecodes.github.io

06 · PROJECT HISTORY — NOON.COM (FEB 2025 – PRESENT)

Noon.com is MENA's largest e-commerce marketplace operating in Saudi Arabia, UAE, Egypt, and GCC markets. Abhinav supports the KSA Beauty and AE/SA FMCG categories with a dataset spanning 34+ tables across sales, assortment, OOS, inventory, competitive pricing, and CRM domains.

■ 06A · Data Engineering · OOS & Pricing

OOS & Pricing Dashboard Engineering

Objective

Build two production pipeline variants powering the KSA Beauty category's core Out-of-Stock (OOS) and Pricing Competitiveness metrics. The dashboard surfaces daily operational health across all major beauty sub-categories and enables account managers to take immediate restocking and pricing action.

Methodology

- Platform OOS Variant — Combined warehouse view; OOS flag considers all fulfilment nodes for full platform visibility.
- Core-Only OOS Variant — Core warehouse only; quantity-to-order formula includes rapid-delivery demand because the express network is fed from core inventory.
- Live Assortment Coverage — Extended via UNION across multiple catalogue sources to capture the full SKU universe visible to customers.
- Fan-Out Prevention — Grain contracts enforced at every join; DISTINCT applied in target-universe CTEs before any multi-grain join.
- NULL-Safe Logic — IS DISTINCT FROM used for partner exclusion to safely handle NULL-valued partner IDs, preventing silent row drops.

Tools & Tech

BigQuery SQL	Airflow DAGs	Inventory & Assortment Tables	Looker Studio

Business Impact

✓ OOS % of GMV Loss maintained below 25% across all 4 beauty sub-categories through Q2 2026.

✓ Twice-daily automated refresh — zero manual analyst intervention for daily operational reporting.

✓ 12 account managers receiving real-time availability and pricing signals to drive restocking decisions.

■ 06B · Dashboards · Seller Analytics

AM Seller Performance Dashboard

Objective

Build a production Airflow pipeline tracking account-manager-level seller performance across KSA Beauty, running twice daily. Outputs four tables at different business grains enabling daily performance management across all 12 account managers.

Methodology

- Four output grains: Seller-level (GMV, Units, GVs, OOS%, PC%), Brand-level (GMV, Units, ASP, YoY), Key-Account (L'Oreal portfolio special mapping), and Category Manager rollup (targets, attainment%).
- AM-to-seller mapping applied at paired category+seller grain — prevents cross-AM seller attribution leakage when a seller operates across multiple sub-categories.
- Brand-specific conditional logic for key strategic brands ensuring correct attribution under the relevant account manager.

• Deep-dive pipeline runs daily and produces four analytical outputs: Overall, Electronics Personal Care, Personal Styling Tools, and Brand grains.

Tools & Tech

BigQuery SQL	Airflow TOML DAGs	Sales & Assortment Datasets	Looker Studio

Business Impact

✓ 12 account managers receive a twice-daily source of truth — eliminating ad-hoc manual pulls.

✓ Category manager view connects commercial actions to category P&L; attainment weekly.

✓ Performance conversations shifted from post-mortem to proactive mid-week intervention.

■ 06C · CRM · Customer Analytics

WhatsApp CRM Win-Back Cohort Engine

Objective

Build two distinct customer cohorts for a WhatsApp CRM retargeting campaign. Identify high-value potential buyers who showed strong purchase intent but did not convert, and customers who purchased competitor products and can be won to the flagship brand.

Methodology

• Cohort 1 — High-Intent Non-Purchasers: mined raw clickstream events to find customers with product-detail-page views on target SKUs within a 60-day window, then excluded anyone who ultimately purchased.

• Cohort 2 — Competitor Switchers: joined the sales base to the product catalogue to find buyers of competitor devices in a 12-to-6-month window (long enough to be open to an upgrade), excluding existing brand owners.

• Customer dimension deduplication using latest-non-null field resolution to resolve multiple PII records per customer.

• Cohort precision enforced at every step — dirty cohorts waste CRM channel budget.

Tools & Tech

BigQuery SQL	Clickstream Event Tables	Sales Base	Customer Dimension (PII)
ARRAY_AGG deduplication			

Business Impact

✓ Campaign-ready cohorts where every recipient had a data-backed reason to receive the message.

✓ Methodology became a reusable template for future win-back and cross-sell campaigns across categories.

✓ Precision targeting protecting the WhatsApp channel from spam fatigue while maximising conversion potential.

■ 06D · Brand Analytics · Beauty

L'Oreal 10-Brand YoY SKU-Level Scorecard

Objective

Produce a comprehensive year-on-year SKU-level performance scorecard for a major global beauty brand's portfolio on the platform, covering 10 brands across both retail (SI) and marketplace (NSI) channels, used for quarterly business reviews.

Methodology

• Three data domains joined at SKU grain: sales fact base, product-view analytics, and search funnel impression tables.

• All dimensional attributes sourced from the product catalogue (never from transaction tables where they drift).

• SKU universe deduplicated before joins to prevent fan-out — grain contracts enforced throughout.

• Search-intent demand (customers searching for the brand) separated from discovery demand (platform-introduced) — a distinction that changes marketing investment conversations.

• Date-sharded table logic hardened with explicit operator precedence after catching a subtle filter bug.

Tools & Tech

BigQuery SQL	Date-Sharded Tables	Search Funnel Analytics	YoY Comparison Logic
Google Sheets (QBR delivery)			

Business Impact

✓ Armed the category team with a defensible SKU-deep narrative for one of the world's largest beauty groups.

✓ Caught a silent filter bug before it distorted a major brand investment decision.

✓ Directly supported joint business planning and marketing investment discussions with the brand partner.

■ 06E · Executive Analytics · CEO-level

IPL Category Deep-Dive & CEO Deck (~190K Customers)

Objective

Build a comprehensive category deep-dive into the IPL (Intense Pulsed Light) hair removal segment covering the full L1–L4 funnel. Delivered as both an AM-facing deep dive and a CEO-level 7-tab report with a production pipeline.

Methodology

• Full L1-L4 funnel: Beauty category → Electronics Personal Care → Personal Styling Tools → Brand level.

• ~190,000-customer cohort: Month-on-month new customer tracking using each customer's first-ever category purchase date.

• Cross-category affinity mapping: what these customers buy before, with, and after the device (same-basket detection + time-from-first-purchase analysis).

• Critical fan-out bug identified and fixed: joining target universe at SKU grain without DISTINCT caused 4x GMV inflation — self-caught before any output was shared.

• Two production pipeline outputs: customer cohort sizing (Part A) + post-purchase cross-category affinity (Part B).

• Excel deep-dive: dynamic brand selector dropdown, live formula connections from raw data tabs, per-brand metric derivations and MoM trend charts.

Tools & Tech

BigQuery SQL	Airflow DAG	Excel (Multi-tab, Dynamic)	Funnel Analytics
Cohort & Basket Analysis			

Business Impact

✓ Quantified answer to 'is this category a customer-acquisition engine?' — with evidence on cross-category spend timing.

✓ Affinity findings informed bundling decisions, CRM journeys, and assortment bets.

✓ ~190K customer acquisition tracker became the standing measure of category health at executive level.

■ 06F · Commercial Analytics · GCC

GCC Cross-Border GMV Tracker

Objective

Track GMV from GCC cross-border orders (customers in Oman, Kuwait, Bahrain, Qatar purchasing from the platform's primary catalogues) on a weekly basis to identify demand spikes and guide inventory allocation.

Methodology

- Four GCC markets tracked beyond core markets: Oman, Kuwait, Bahrain, Qatar.
- Grain: SKU x Country x Week, rolled to Category x Country x Week for management view.
- Metrics: GMV, Units, Orders, ASP, Week-on-Week delta.
- Scheduled query with auto-append to master tracking table for continuity.

Tools & Tech

BigQuery SQL	Looker Studio	Scheduled Queries	

Business Impact

✓ Identified cross-border white-space opportunities used in supplier negotiations.

✓ Category managers can proactively allocate inventory ahead of cross-border demand spikes.

■ 06G · Planning · FMCG

FMCG AOP 53-Week Tracker (AE/SA)

Objective

Build a 53-week Annual Operating Plan tracker for FMCG categories in both UAE and Saudi Arabia, enabling Category Managers to monitor weekly attainment against annual targets across Grocery, Baby Consumables, and Hair & Personal Care.

Methodology

- Long-format SQL (week x market x sub-category) — single table for Looker compatibility, not pivoted.
- 53 weeks to handle ISO week calendar edge cases in fiscal year planning.
- Both AE and SA markets in separate rows, joinable on week + sub-category.
- POOS flag (Days of Inventory < 30) added for inventory health signalling alongside AOP metrics.
- Sub-category dropdown in Sheets layer for AM-level drill-down without modifying SQL.

Tools & Tech

BigQuery SQL	Google Sheets (AOP template)	Looker Studio	

Business Impact

✓ Category Managers track weekly attainment vs. annual target in one view for two markets.

✓ Inventory health (POOS) integrated with financial planning — connecting stock risk to revenue risk.

■ 06H · Campaign Analytics · Attribution

NTB Cohort & Coupon Campaign Attribution

Objective

Measure new-to-brand (NTB) customer acquisition via a major coupon campaign. Identify how many redemptions resulted in a customer's first-ever brand purchase on the platform and track subsequent repeat behaviour — proving long-term campaign value beyond first-purchase GMV.

Methodology

- NTB definition: customer's first-ever purchase of the brand on the platform (lifetime lookback).
- Cohort splits: New to Platform vs. Existing customer; new-to-grocery sub-analysis included.
- Critical bug fixed: cross-date JOIN inflation — joining without a date-range boundary caused 4x row multiplication; fixed by scoping JOIN to campaign window only.

• 30/60/90-day repeat purchase rate tracked for NTB vs. existing customer cohorts.

Tools & Tech

BigQuery SQL	Sales Base	Google Sheets	

Business Impact

✓ Campaign-level NTB%, repeat rate, and GMV per cohort delivered in CEO summary slide.

✓ Bug fix prevented a significant overstatement of campaign ROI from being reported to senior leadership.

✓ Cohort methodology now reused for all future coupon attribution analyses.

■ 06I · Customer Analytics · Demographics

Customer Demographics Analysis

Objective

Build a 5-year demographic profile of hair removal device buyers on the platform covering age, gender, nationality, and city distribution. Used by brand teams and Category Managers to inform media targeting, assortment decisions, and campaign localisation.

Methodology

• PII join: linked order-level data to customer demographic table via customer identifier.

• ARRAY_AGG deduplication: resolved multiple PII records per customer using latest-non-null field resolution (ordered by last-updated timestamp) to prevent fan-out.

• Dimensions: Age bands (18-24, 25-34, 35-44, 45+), Gender, Nationality (top 15), City.

• 5-year lookback with YoY cohort breakdown.

• Output: demographic heatmap in Sheets; pie charts for gender/nationality; bar charts for age/city.

Tools & Tech

BigQuery SQL	ARRAY_AGG Deduplication	Customer PII Table	Looker Studio
Google Sheets			

Business Impact

✓ Brand team received a full demographic picture enabling media targeting and campaign localisation.

✓ Deduplication pattern now a standard template across all customer-enrichment queries.

■ 06J · Financial Analytics · AP/AR

PO-Level AP/AR Aging Framework

Objective

Build a four-snapshot monthly AP (Accounts Payable) and AR (Accounts Receivable) aging framework at Purchase Order level for the beauty category, giving Category Managers and Finance visibility into how long outstanding payables and receivables have been ageing across supplier accounts.

Methodology

• Four monthly snapshots enabling month-on-month aging trend tracking.

• Granularity: PO x Supplier x Month — lowest actionable level for finance follow-up.

• Aging buckets: 0-30, 31-60, 61-90, 90+ days.

• AP view (outstanding payables to suppliers) and AR view (receivables from marketplace sellers) in one framework.

• Finance-ready Excel with color-coded aging buckets and escalation flags for 90+ day items.

Tools & Tech

BigQuery SQL	Google Sheets	Excel	Finance Tables

Business Impact

✓ Finance team moved from quarterly manual aging reports to monthly automated snapshots.

✓ Escalation flags for 90+ day items enabling proactive supplier follow-up and cash flow management.

■ 06K · Operations Analytics · Per-AM

Per-AM OOS & Price Competitiveness Query Suite

Objective

Build a complete query suite for OOS and Price Competitiveness metrics tailored to each of the 12 Account Managers. Each AM owns specific sellers across specific sub-categories, and queries must respect those boundaries for attribution accuracy.

Methodology

- Monthly managed seller list as canonical AM mapping — point-in-time snapshot preventing retro-attribution errors.
- Sub-category + Seller ID paired filtering — prevents cross-AM seller leakage when a seller operates in multiple sub-categories.
- LEFT JOIN always for the product catalogue — ensures all listed SKUs appear even if no sales in period.
- OOS metric grain: daily ratio first, then average rollup to monthly — mathematically correct operational visibility.
- PC metric: competitive price count divided by live price count per day, averaged to monthly.

Tools & Tech

BigQuery SQL	Competitive Pricing Dataset	Inventory Tables	Looker Studio

Business Impact

✓ 12 AMs each with a personalised, attribution-accurate view of their book of business.

✓ Pricing and availability conversations with sellers grounded in correct, boundary-respecting data.

■ 06L · Assortment Analytics · Seller Insights

Seller Demographics & Assortment Analysis

Objective

Analyse the platform's seller landscape within the Beauty category focusing on a specific seller cohort. Build a 5-year demographic profile of their buyers and assortment footprint analysis.

Methodology

- Seller list filtered to Beauty-relevant rows from a larger cross-category dataset.
- Buyer demographics: age, gender, nationality, city — 5-year lookback matching the standard deduplication pattern.
- Assortment analysis: UNION of core and express catalogue at SKU x date level (in-stock only), rolled to month via COUNT DISTINCT SKU.
- Key insight: buyer base skews younger (25-34) and female — informing assortment and marketing recommendations.

Tools & Tech

BigQuery SQL	Customer PII Tables	Assortment Tables	Looker Studio

Business Impact

✓ Output used for Beauty assortment planning and AM seller onboarding decisions.

✓ Demographic insight enabled targeted marketing recommendations for the seller cohort.

■ 06M · Operations · Inventory Quality

Retail Problem-Stock Analysis

Objective

Analyse a 14-sheet Excel workbook representing a warehouse problem-stock aging tracker. Identify high-risk SKUs, flag data quality issues, and provide actionable prioritisation for the warehouse operations team.

Methodology

- 14 sheets covering different warehouse locations and product categories.
- Six time buckets: A (0-7 days), B (8-14), C (15-30), D (31-60), E (61-90), F (90+ days).
- Value concentration finding: majority of GMV-at-risk concentrated in high-risk 60+ day buckets (E+F).
- Data quality flags: broken embedded charts, blank legacy columns, case-mismatch duplicate BU label.
- Output: cleaned Excel with corrected labels, removed legacy columns, rebuilt pivot summary.

Tools & Tech

Excel (Multi-sheet)	Pivot Tables	Data Quality Assessment	

Business Impact

✓ Recommended focus on liquidating E+F buckets — highest GMV recovery potential.

✓ Data quality issues documented and corrected, enabling reliable week-on-week bucket migration tracking.

07 · PROJECT HISTORY — FREELANCE ANALYTICS (NOV 2022 – FEB 2025)

Multi-client engagements across FMCG, e-commerce, fintech, and healthcare. Deepened BigQuery, Looker Studio, and Python automation capabilities across diverse business contexts.

Project Type	Domain	Tools	Key Output
Retention Cohort Dashboard	E-commerce / D2C	BigQuery, Looker Studio	Monthly cohort retention rates; churned customer ale
Sales Funnel Automation	FMCG / CPG	BigQuery, Apps Script, Sheets	Automated weekly funnel report; 6-stage drop-off an
Revenue Attribution Model	Fintech / Payments	Python, SQL, Excel	Multi-touch attribution; LTV modelling
Inventory Health Dashboard	Retail / E-commerce	BigQuery, Looker Studio	OOS%, POOS, DOH tracker; weekly automated refr
Customer Segmentation (RFM)	E-commerce	Python (pandas, sklearn)	RFM segmentation; 5 customer tiers; campaign lists
Financial Reporting Pipeline	SME / Services	Google Sheets, Apps Script	Automated P&L, cash flow; 80% manual time reduct
Social Media Analytics Tool	Marketing Agency	Python, Sheets	Cross-platform engagement tracker; content scoring

08 · PROJECT HISTORY — EUCLOID (JAN–OCT 2022)

Role: Senior Product Analyst | B2B SaaS / Product Analytics

Eucloid is a B2B data analytics company. As Senior Product Analyst, Abhinav owned funnel analytics, retention cohort modelling, and A/B testing infrastructure for product decisions.

Funnel Analytics

L1-L4 product funnel for SaaS onboarding — identified 3 key drop-off points reducing trial-to-paid conversion.

Retention Cohort Modelling

Weekly cohort tables for Day-1, Day-7, Day-30 retention; surfaced 35% Day-30 retention vs 50% industry benchmark.

A/B Testing Infrastructure

Designed experiment tracking schema; built statistical significance calculator in Python (scipy stats).

Event Tracking Framework

Defined 40+ custom events with engineering team; Mixpanel / Amplitude dashboards for product decisions.

MAU / DAU Reporting

Automated via scheduled queries feeding Looker dashboards; engagement 20% up, conversion 25% up.

Churn Prediction Model

Logistic regression on 6-month feature window; deployed as weekly alert for customer success team.

`Tools: Python (pandas, sklearn, scipy) · BigQuery · Mixpanel · Amplitude · Looker · Excel`

09 · PROJECT HISTORY — UNITEDHEALTH GROUP (AUG 2020 – JAN 2022)

Role: Data Scientist | Healthcare Analytics & Machine Learning

Claims Cost Prediction Model

Predicted 12-month claims cost per member using historical claims, diagnosis codes, and demographic features. XGBoost model with 150+ engineered features (AUC: 0.84). Deployed as monthly batch scoring job enabling actuarial team to refine premium pricing for high-risk segments.

NLP Document Classification

Classified clinical notes and pre-authorisation documents into 15 categories using spaCy + sklearn pipeline. Reduced manual review workload by 60% for the clinical operations team.

Risk Stratification Dashboard

Tableau dashboard stratifying member population into 5 risk tiers based on claims history, comorbidity index, and engagement score. Used by case managers for proactive outreach prioritisation.

Fraud Detection POC

Anomaly detection on claims patterns using Isolation Forest. Flagged 120+ suspicious billing patterns for the Special Investigations Unit.

ETL Pipeline Refactoring

Automated Dialer reporting cutting manual effort 80%; refactored ETL pipelines for 30% efficiency gain; reduced oncology workload 70%.

`Tools: Python (XGBoost, sklearn, spaCy, pandas) · SQL · Tableau · Azure ML · Talend ETL`

10 · PROJECT HISTORY — CITIBANK (JUL 2018 – AUG 2020)

Role: Business Analyst | Credit Analytics & Regulatory Reporting | Bengaluru

First role post-IIT Kanpur. Citibank's Bengaluru centre is a global capabilities hub supporting risk, finance, and analytics for Citi's Asia-Pacific and EMEA regions.

Credit Portfolio Analytics

Monthly credit portfolio health reports covering DPD buckets, NPA ratios, and provision coverage across retail and corporate segments. Built in SAS + Excel with automated refresh via SAS macros.

Regulatory Reporting (Basel III)

Supported quarterly Basel III capital adequacy reporting — RWA calculations, Tier 1/2 capital ratios, liquidity coverage ratio — for submission to RBI and internal risk committees.

Customer Segmentation & Revenue

K-means clustering (Python/SQL) isolating top customers driving 84% of sales; 24-month inactivity closure rule preserving ~100M in potential revenue.

Customer Attrition Model

Logistic regression predicting credit card attrition risk using 90-day behavioral features. Used by retention team for proactive campaign targeting.

KPI Dashboard for Risk Committee

Monthly Excel KPI dashboard presented to Asia-Pacific Risk Committee. Reduced manual compilation from 2 days to 3 hours. Metrics: NPA%, Coverage Ratio, Slippage Rate, Credit Cost.

`Tools: SAS · SQL · Excel (VBA/Macros) · PowerPoint · Tableau · Python`

11 · AGENTIC AI & AUTOMATION PROJECTS

LinkedIn Auto-Poster — The 1% Insight

Fully deployed Node.js + Anthropic Claude API system on Railway. Posts twice daily to LinkedIn. OAuth2 token management with 60-day refresh cycle. Posts tailored by time slot (analytical AM, viral PM). GitHub: github.com/agdecodes

Portfolio Website — agdecodes.github.io

Single-file HTML portfolio with dark premium theme, purple-to-amber gradient branding, interactive project case study modals, SQL terminal animation, and fully responsive layout. Hosted on GitHub Pages.

Visual Work History PDF Engine

Multi-chart PDF with status donut, category bar, tool frequency histogram, project timeline, and OOS health tracker — built entirely in Python matplotlib without any manual design tool.

LangGraph / LangChain Exploration

Learning agentic AI frameworks: Hugging Face AI Agents Course, LangGraph. Building toward multi-agent analytics workflows that automate end-to-end data storytelling.

12 · CAREER GOALS & VISION 2026–2028

2026 Goals

- Team Leadership: Onboard and manage junior analysts — first step toward a people-management track.
- Reporting Automation: Migrate all recurring manual reports to fully automated pipelines; reduce manual touch by 90%.
- AI Integration: Deploy at least one agentic AI workflow into production analytics work.
- Personal Brand: Grow The 1% Insight via consistent automated content.
- Portfolio: Maintain agdecodes.github.io as a living proof-of-work document.

2027–2028 Vision

- Target role: Category Analytics Manager or Head of Analytics at a MENA / India e-commerce or FMCG company.
- Team size: Build and lead a team of 4–6 analysts across markets.
- AI-native analytics: Pioneer an AI-native analytics function using LLM-assisted querying and agentic reporting workflows.
- Consulting: Begin selective fractional consulting under the AG Analytics brand — serving mid-market FMCG and e-commerce clients.
- Thought leadership: Conference talks and LinkedIn Live on e-commerce analytics, AI in BI, and data career building.

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Abhinav Goyal (AG) · agdecodes.github.io · avi290994@gmail.com